BIRATIONAL MORPHISMS OF DECENT SPACES ARE ISOMORPHIC OVER THE GENERIC POINTS OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5256-5266, immediately following the statement at 5246-5254.

Editorial note. Source line 5261 reads 'is consists of points of codimension 0' (grammar slip in the original); retained verbatim.

Editorial note. One cross-chapter pointer (Morphisms) and two same-chapter line-indexed stubs are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Maximal representable open subspace. Lemma lemma-finitely-many-irreducible-components of this chapter, cited at proof lines 5257-5259. Supplies the maximal open subspaces X' subset X and Y' subset Y which are representable (schemes), using finitely many irreducible components.
- Generic fibres land in codimension zero. Lemma lemma-quasi-finiteness-over-generic-point of this chapter, cited at proof lines 5260-5262. The fibre of f over the generic point y consists of codimension-0 points of X, hence lies inside X' (generic points survive representable restriction).
- Scheme-side birational generic fibre isomorphism. Morphisms, Lemma morphisms-lemma-birational-generic-fibres, cited at proof lines 5265-5266. The imported terminal isomorphism for schemes.

Source reference. Original lemma 0BB9: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4911–4911. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BBB: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5085–5085. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAB: focused-seeds/geometry/S9-decent-birational-generic-fibres/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a birational morphism of algebraic spaces over S which are decent and have finitely many irreducible components. If $y \in Y$ is the generic point of an irreducible component, then the base change $X \times_Y \mathrm{Spec}(\mathcal{O}_{Y,y}) \to \mathrm{Spec}(\mathcal{O}_{Y,y})$ is an isomorphism.*

Proof. Let $X' \subset X$ and $Y' \subset Y$ be the maximal open subspaces which are representable, see Lemma 0BB9. By Lemma 0BBB the fibre of f over y is consists of points of codimension 0 of X and is therefore contained in X'. Hence $X \times_Y \mathrm{Spec}(\mathcal{O}_{Y,y}) = X' \times_{Y'} \mathrm{Spec}(\mathcal{O}_{Y',y})$ and the result follows from Morphisms, Lemma 0BAB. □